

# Roman Ibrahimov

## CONTACT INFORMATION

ARG & Embodied Computation Lab  
Princeton University, Princeton, NJ 08544  
**U.S. Permanent Resident**

*E-mail:* [roman.ibrahimov@princeton.edu](mailto:roman.ibrahimov@princeton.edu)  
*Webpage:* [roman-ibr.github.io](http://roman-ibr.github.io)

## EDUCATION AND PROFESSIONAL DEVELOPMENT

**University of California, Berkeley, CA, USA**  
**M.S.**, Mechanical Engineering  
*Advisor: Prof. Mark Müeller*

**Purdue University, West Lafayette, IN, USA**  
**M.S.**, Aeronautics and Astronautics  
*Advisors: Prof. Shirley Dyke and Prof. David Cappelleri*

- **IEEE RAS Summer School on Multi-Robot Systems**, CTU, Prague, Czechia  
*Multi-UAV control, perception, localization, and planning*
- **DroneCamp**, University of California, ANR, Monterey, California, USA  
*UAV hardware & sensors, mission planning, flight skills, safety & regulations*

**Skoltech** (in collaboration with **MIT**), Moscow, Russia  
**M.S. with Distinction**, Space and Engineering Systems  
*Advisor: Prof. Dzimitry Tsetserukou*

**ADA University, Baku, Azerbaijan**  
**B.S. Summa Cum Laude**, IT and Systems Engineering

- **ITMO University, St. Petersburg, Russia**,  
*Exchange Student, Control Systems and Robotics*
- **Middle East Technical University, Ankara, Turkey**,  
*Exchange Student, Electrical and Electronics Engineering*

## RESEARCH INTERESTS

*Planning & Control; Robot Learning; Contact-Rich Manipulation; Autonomous Systems; Cyber-Physical Systems;*

## PUBLICATIONS

**R. Ibrahimov**, T. Yang, and M. W. Müeller, “Kalman Filter-Based Drift Detection and Mitigation of Visual-Inertial Odometry in UAVs,” *American Control Conference (ACC)*, 2025.

T. Yang, **R. Ibrahimov**, and M. W. Müeller, “Towards Safe and Efficient Through-the-Canopy Autonomous Fruit Counting with UAVs,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

**R. Ibrahimov** and Jannik Heinen., “Multi-Robot Distributed Optimization for Exploration and Mapping of Unknown Environments using Bioinspired Tactile-Sensor,” *arXiv preprint*, 2025.

A. Behjat, **R. Ibrahimov**, A. Lenjani, A. Barket, K. Martinus, A. Maghareh, D. Whitaker, I. Bilonis, and S. Dyke, “A Computational Framework for the Evaluation of Resilience in Deep Space Habitat Systems,” *ASME 2022 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference*, St. Louis, Missouri, USA, Aug. 14-17, 2022.

E. Tsykunov, A. Fedoseev, E. Dorzhieva, R. Agishev, **R. Ibrahimov**, D. Vasquez, L. Labazanov, and D. Tsetserukou, “DroneStick: Flying Joystick as a Novel Type of Interface,” *ACM SIGGRAPH Asia 2021 Emerging Technologies*, Virtual, Dec. 14-17, 2021.

E. Karmanova, V. Serpiva, S. Perminov, **R. Ibrahimov**, A. Fedoseev, and D. Tsetserukou, “SwarmPlay: A Swarm of Nano-Quadcopters Playing Tic-tac-toe Board Game against a Human,” *ACM SIGGRAPH 2021 Emerging Technologies*, Virtual, Aug. 9-13, 2021.

**R. Ibrahimov**, N. Zherdev, and D. Tsetserukou, “DroneLight: Drone Draws in the Air using Long Exposure Light Painting and ML,” *29th IEEE International Conference on Robot & Human Interactive Communication (IEEE RO-MAN 2020)*, Naples, Italy, Aug. 31-Sept. 4, 2020.

E. Tsykunov, R. Agishev, **R. Ibrahimov**, T. Moriyama, L. Labazanova, H. Kajimoto, and D. Tsetserukou “SwarmCloak: Landing of Two Micro-Quadrotors on Human Hands Using Wearable Tactile Interface Driven by Light Intensity,” 2020 IEEE Haptics Symposium, Washington DC, USA, March 28-31, 2020.

**R. Ibrahimov**, E.Tsykunov, V. Shirokun, A. Somov, and D. Tsetserukou, “DronePick: Object Picking and Delivery Teleoperation with a Drone Controlled by a Tactile Wearable,” 28th IEEE International Conference on Robot & Human Interactive Communication (IEEE RO-MAN 2019), New Delhi, India, 2019.

E. Tsykunov\*, **R. Ibrahimov\***, D. Vasquez, D. Tsetserukou, “SlingDrone: System for Navigation and Interaction with the Environment Using a Single Drone and VR,” 25th ACM Symposium on Virtual Reality Software and Technology (VRST 2019), Sydney, Australia, 2019.

\*- authors contributed equally to the paper.

E. Tsykunov, R. Agishev, **R. Ibrahimov**, L. Labazanova, T. Moriyama, H. Kajimoto, D. Tsetserukou, “Swarm-Cloak: Landing of a Swarm of Nano-Quadrotors on Human Arms,” Int. Conf. on Computer Graphics and Interactive Technologies (ACM SIGGRAPH Asia 2019), Emerging Technologies, Brisbane, Australia, 2019.

E.Tsykunov, R. Agishev, **R. Ibrahimov**, L. Labazanova, A. Tleugazy, and D. Tsetserukou, “SwarmTouch: Guiding Swarm of Nano-Quadrotors with Impedance Control using Wearable Tactile Interface,” IEEE Transactions on Haptics, 2019.

G. Yashin, D. Trinitatova, R. Agishev, **R. Ibrahimov**, and D. Tsetserukou, “AeroVR: Virtual Reality Teleoperation System for the UAV Robotic Manipulator,” 19th IEEE International Conference on Advanced Robotics (ICAR 2019), Belo Horizonte, Brazil, 2019.

## PROJECT EXPERIENCE

**Contact-Rich Robotic Manipulation for Construction**, Princeton University *July 2025-present*

- Developing a contact-rich manipulation framework for automated assembly tasks using a UR20 robot
- Generating large-scale synthetic demonstration datasets in Isaac Sim for robust sim-to-real transfer
- Training and evaluating diffusion policy augmented with adaptive control for zero-shot real-world performance

**Autonomous Drone Inspection of Caltrans Bridges**, UC Berkeley *Aug 2023-May 2025*

- Implemented visual-inertial odometry with OpenVINS, integrating IMU and stereo camera
- Configured and tested hardware, including RealSense D455 for depth imaging, Qualcomm RB5 for onboard computing and Pixracer R15 for flight control
- Managed system communication via the Robot Operating System (ROS)

**An inspection robot for Duke Energy Power Lines**, Purdue University *May 2022-Aug 2023*

- Implemented Model Predictive Control (MPC) for trajectory tracking along power lines
- Developed line detection algorithms using OpenCV for enhanced visual recognition
- Modeled and simulated robot dynamics
- Configured and tested Argo rover for optimal field deployment and reliability

**Resilient Extraterrestrial Habitats**, NASA RETH Institute, Purdue University *May 2021-Aug 2023*

- Designed a control-theoretic autonomy framework to support resilient design and operation
- Built automated active learning framework with robots and humans-in-the-loop
- Developed methods for detection and diagnosis of anticipated and unanticipated faults
- Contributed to establishing SmartHabs with autonomous abilities to sense, anticipate and respond

**Bio-inspired nano-quadcopter for map building**, Purdue University *Sep 2021-May 2022*

- Developed an API on FreeRTOS to read sensor reading on the quadcopter
- Transmitted sensor data to the ROS base station via radio communication
- Collected and processed point cloud data to construct detailed environmental maps
- Applied machine learning techniques to predict the map of the environment

**Human-Drone Interaction through a Tactile Wearable**, Skoltech *Oct 2018-May 2020*

- Implemented human-drone communication using impedance control and vibrotactile feedback
- Designed a tactile wearable equipped with eccentric rotating mass (ERM) motors for intuitive interaction
- Created a Virtual Reality (VR) application in C# to support teleoperation and aerial manipulation tasks
- Developed remote object manipulation techniques using drones

**Balloon Satellite for Testing Solar Cells in High Altitude**, Skoltech *Sep 2018-Jan 2019*

- Applied Model-based Systems Engineering principles to ensure mission success
- Constructed a CubeSat featuring an onboard controller, solar cells, storage devices, and GPS tracking
- Gathered and analyzed current, voltage, and temperature (CVT) data from solar cell tests

- Retrieved payload with no damage after landing from 35km maximum altitude

**Gas Leak Detecting Mobile Robot for NICA Collider, JINR**

*Summer 2018*

- Designed autonomous navigation system around elliptical collider
  - Built a mobile robot with an on-board temperature camera
  - Developed Computer Vision (CV) algorithm based on Python for detection nitrogen gas leak from the collider
  - Came up with computer-based user interface for remote monitoring
- 

**WORK  
EXPERIENCE**

**Robotics Research Associate, Princeton University, NJ, USA**

*Jul 2025–present*

- Develop and evaluate robotic manipulation strategies for contact-rich construction tasks using UR20 manipulators.
- Collect and analyze multimodal datasets from real-life demonstrations and synthetic simulation environments.
- Implement ROS 2 control frameworks integrating force/torque feedback for precise alignment and insertion.
- Collaborate with faculty and graduate students on experimental design, system integration, and publication efforts.

**Graduate Student Instructor, UC Berkeley, CA, USA**

*Jan 2025-May 2025*

- ENGIN07 Computer Programming for Scientists and Engineers
- Led coding labs and exercises, supporting course delivery.
- Conducted hands-on sessions to apply concepts in programming tasks.
- Held office hours to answer questions and assist with assignments.
- Designed and graded coding assignments, providing feedback on student work.

**Graduate Research Assistant, UC Berkeley, CA, USA**

*Aug 2023-May 2025*

- Design and test control algorithms for UAV stability and obstacle avoidance in complex environments.
- Optimize data processing workflows, enhancing efficiency and accuracy for real-time inspection feedback.
- Collaborate with interdisciplinary teams to troubleshoot hardware and software challenges.
- Presented research findings within lab meetings and contribute to technical publications on UAV autonomy.

**Graduate Research Assistant, NASA RETH Institute, IN, USA**

*May 2021-Aug 2023*

- Developed real-time fault detection algorithms and situational awareness systems to enhance safety and reliability in extreme environments.
- Designed and configured hardware systems, including specialized sensors, actuators, and computational units, to support autonomous operation in space habitats.
- Conducted simulations and tests to validate habitat resilience against environmental and operational disruptions.
- Collaborated with interdisciplinary teams to integrate control-theoretic approaches into habitat design, advancing NASA's SmartHab initiative with enhanced autonomous capabilities.

**Teaching Assistant, Purdue University**

*Jan 2021-May 2022*

- CNIT 155 Introduction to Software Development Concepts (in Python)
- Supported course delivery by leading coding labs and exercises, guiding students in foundational Python programming and problem-solving.
- Conducted hands-on lab sessions, helping students implement concepts through coding exercises and real-world applications.
- Held office hours to address individual student questions, clarify complex topics, and provide assignment assistance.
- Created and evaluated coding assignments, offering constructive feedback to enhance students' programming skills and understanding of software development principles.

**Part-time Instructor, ILC School in Barcelona S.L, Barcelona, Spain**

*June 2020-Present*

- Taught SAT math and GRE quantitative reasoning to high-school and undergraduate students.
- Mentored engineering undergraduates and supervised their data-science and web-application projects.
- Guided students through research poster preparations for ACM WomENCourage.
- Assisted underrepresented students voluntarily with instruction and mentorship to promote STEM participation.

**Intern, Universal Robots, Moscow, Russia**

*Summer 2019*

- Assisted in research and development for Human-Robot Collaboration (HRC) applications involving quadcopters to enhance interaction and safety.
- Participated in designing and testing control strategies to improve quadcopter responsiveness and collaboration

capabilities.

- Supported hardware configuration and troubleshooting to ensure seamless integration of HRC functionalities.
- Documented findings and presented insights to the team, contributing to advancements in HRC applications.

**Intern, Azercosmos OJSCo, Baku, Azerbaijan**

Summer 2017

- Supported the Ground Control Department in monitoring and maintaining satellite communication systems.
  - Assisted with network configuration and troubleshooting to ensure stable connectivity and optimal system performance.
  - Gained experience in analyzing and managing data flow within ground control networks.
  - Documented operational procedures and reported on system performance, contributing to improved workflow efficiency.
- 

## HONORS AND AWARDS

|                                                                               |               |
|-------------------------------------------------------------------------------|---------------|
| <b>American Control Conference Full Travel Grant \$2000</b>                   | April 2025    |
| <b>Purdue University Poster Symposium Best Poster Award \$750</b>             | April 2022    |
| <b>Skolkovo Foundation Full Graduate Scholarship</b>                          | 2018-2020     |
| <b>ACM SIGGRAPH Asia Best Demonstration Award</b>                             | November 2019 |
| <b>Skolkovo Presidential Scholarship</b>                                      | November 2019 |
| <b>Skolkovo Presidential Travel Award, \$5000</b>                             | November 2019 |
| <b>Skolkovo Presidential Travel Award, \$4500</b>                             | November 2019 |
| <b>President's List of Honor, ADA University</b>                              | 2014-2017     |
| <b>Erasmus+ Exchange Scholarship, METU</b>                                    | Spring 2016   |
| <b>FameLab World Finalist, Cheltenham, the UK</b>                             | June 2015     |
| <b>1st place, Microsoft ImagineCup Competition, Baku, Azerbaijan</b>          | May 2015      |
| <b>4th place, International Rudolf Ortway Competition in Physics, Hungary</b> | Dec 2014      |

---

## MENTORING EXPERIENCE

### **UC Berkeley:**

- Jannik Matthias Heinen, MEng in Mechanical Engineering (upcoming PhD student in TUM, Germany)
- Milan Rosic, MEng in Aerospace Engineering (safety engineer at Lucid Motors)
- Yuxuan Peng, MEng in Mechanical Engineering
- Ruonan Yang, MEng in Mechanical Engineering

### **Purdue University:**

- Anderson Xu, BS in Aero/Astro (PhD student at Purdue University)
  - Benjamin Krugman, BS in Mechanical Engineering
  - Fahad Tajiki, BS in Electrical Engineering (firmware engineer at Saab, Inc.)
- 

## REVIEWER FOR CONFERENCE AND JOURNALS

IEEE Robotics and Automation Letters (RA-L) 2024, RA-L 2023, RA-L 2022, IEEE International Conference on Robotics and Automation (ICRA) 2024, ICRA 2023, ICRA 2021, ICRA 2020; ACM Conference on Human Factors in Computing Systems (CHI) 2020; Virtual Reality & Intelligent Hardware Journal 2020;

---

## TECHNICAL SKILLS

**Programming:** C/C++, Java, Python, MATLAB/Simulink, LabVIEW,  $\text{\LaTeX}$

**Robotics:** Robot Operating System (ROS), ROS2, Gazebo, NVIDIA Isaac Sim, RViz, Unity

**MCUs:** Qualcomm RB5, myRIO, Arduino, Libelium

**Mechanical:** CAD (SolidWorks), 3D Printing, soldering, laser/plasma cutting

---